Common mode filters/Chokes
Automotive signal line (for infotainment)
ACM series

RoHS REACH Halogen Free Lead Free

AEC-Q200

ACM2012 type

Product Portal Search Simulation Model Selection Guide Tech Library Tech Note Application Guides Automotive

FEATURES

- Highly reliable with a temperature range of –40 to +105°C.
- Impedance variation: Extensive lineup are available for compatibility with various usages
- Common mode EMI measure: Possible to suppress waveform common mode EMI without straining the waveform.
- Operating temperature range: -40 to +105°C
- Compliant with AEC-Q200

APPLICATION

- Radiated noise suppression for car multimedia interfaces (MOST, USB2.0, IDB-1394, etc.).

PART NUMBER CONSTRUCTION

ACM	2012	-	900	-	2P	-	T	001
Series name	L×W dimensions 2.0×1.2 mm		Impedance (Ω) at 100MHz		Number of lines		Packaging style	Internal code

CHARACTERISTICS SPECIFICATION TABLE

Common mode impedance [100MHz]* (Ω)min.	(Ω)typ.	DC resistance [1 line] (Ω)max.	Rated current (mA)max.	Rated voltage (V)max.	Insulation resistance (MΩ)min.	Part No.
65	90	0.19	400	50	10	ACM2012-900-2P-T001
90	120	0.22	370	50	10	ACM2012-121-2P-T001
150	200	0.25	350	50	10	ACM2012-201-2P-T001
270	360	0.50	220	50	10	ACM2012-361-2P-T001
510	680	1.3	200	50	10	ACM2012-681-2P-T001
750	1000	1.5	190	50	10	ACM2012-102-2P-T001
1500	2200	2.5	150	50	10	ACM2012-222-2P-T001

Measurement equipment

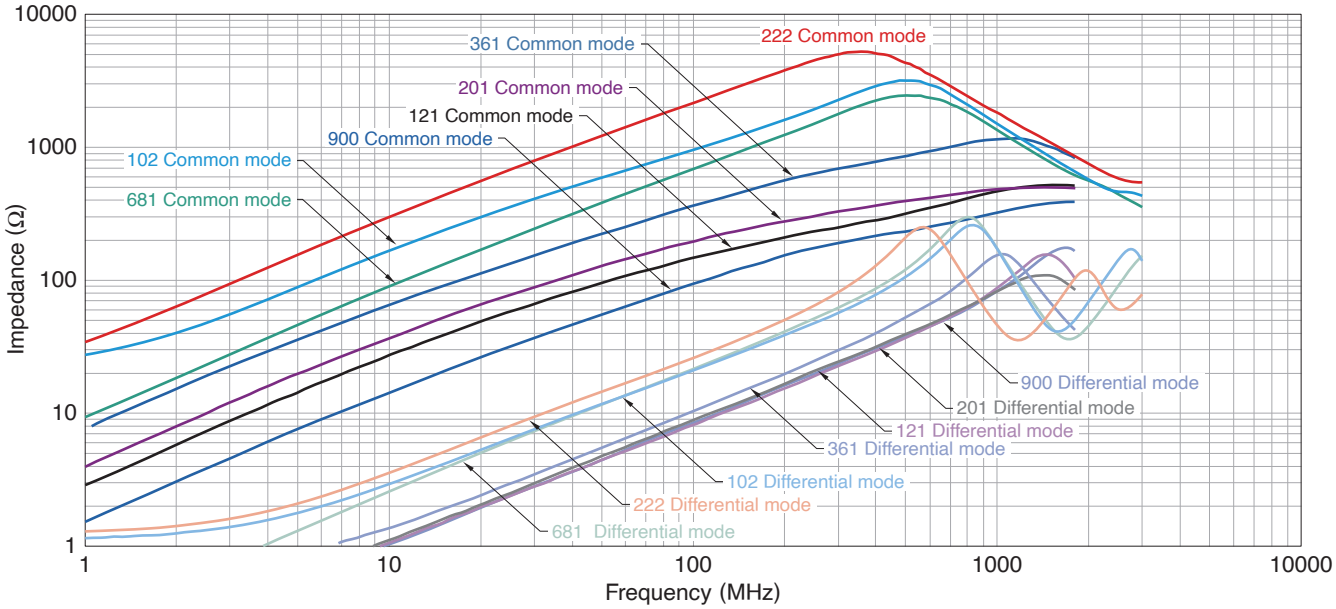
Measurement item	Product No. *	Manufacturer
Common mode impedance	4991A	Keysight Technologies
DC resistance	4338A	Keysight Technologies
Insulation resistance	4339A	Keysight Technologies

* Equivalent measurement equipment may be used.



ACM2012 type

IMPEDANCE VS. FREQUENCY CHARACTERISTICS



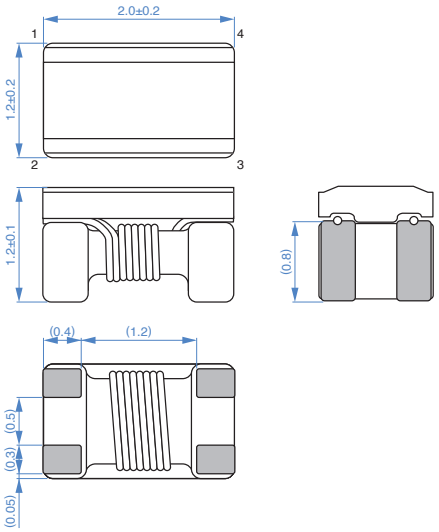
Measurement equipment

Product No. *	Manufacturer
4991A	Keysight Technologies

* Equivalent measurement equipment may be used.

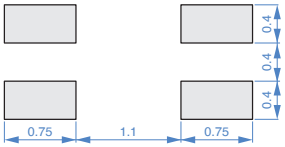
ACM2012 type

SHAPE & DIMENSIONS



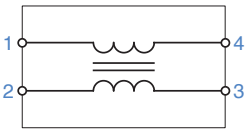
Dimensions in mm

RECOMMENDED LAND PATTERN



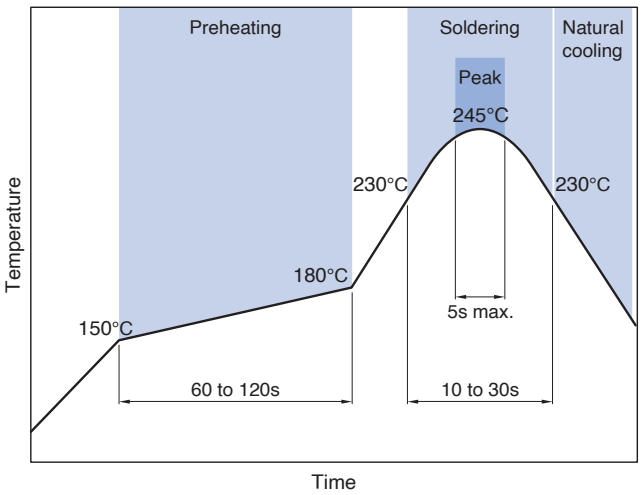
Dimensions in mm

CIRCUIT DIAGRAM



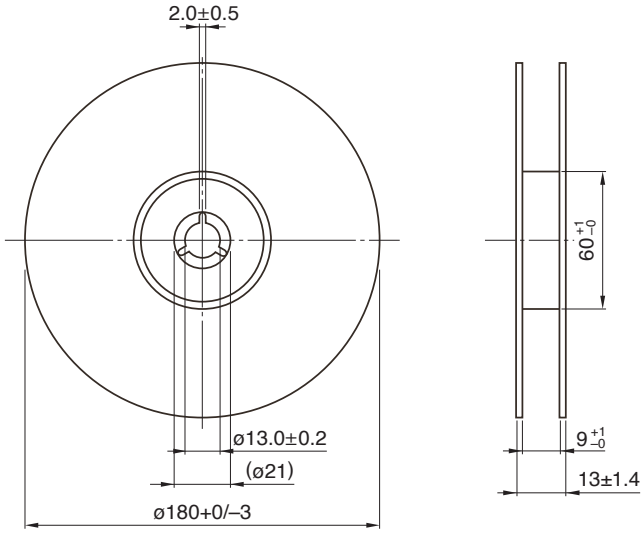
• No polarity

RECOMMENDED REFLOW PROFILE



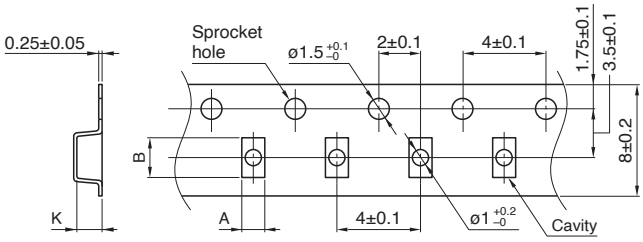
PACKAGING STYLE

REEL DIMENSIONS



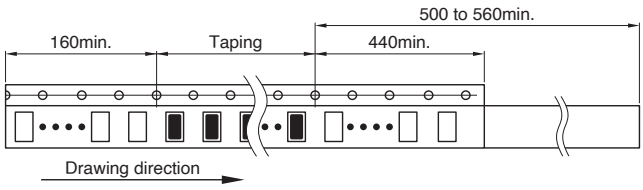
Dimensions in mm

TAPE DIMENSIONS



Dimensions in mm

Type	A	B	K
ACM2012	(1.4)	(2.25)	(1.4)



Dimensions in mm

PACKAGE QUANTITY

Package quantity	2,000 pcs/reel

TEMPERATURE RANGE, INDIVIDUAL WEIGHT

Operating temperature range *	Storage temperature range **	Individual weight
−40 to +105 °C	−40 to +105 °C	10 mg

* Operating temperature range includes self-heating.
** The storage temperature range is for after the assembly.

REMINDERS FOR USING THESE PRODUCTS

Before using these products, be sure to request the delivery specifications.

SAFETY REMINDERS

Please pay sufficient attention to the warnings for safe designing when using this products

REMINDERS

- The storage period is within 12 months. Be sure to follow the storage conditions (temperature: 5 to 40°C, humidity: 10 to 75% RH or less).
  If the storage period elapses, the soldering of the terminal electrodes may deteriorate.
- Do not use or store in locations where there are conditions such as gas corrosion (salt, acid, alkali, etc.).
- Soldering corrections after mounting should be within the range of the conditions determined in the specifications.
  If overheated, a short circuit, performance deterioration, or lifespan shortening may occur.
- When embedding a printed circuit board where a chip is mounted to a set, be sure that residual stress is not given to the chip due to the overall distortion of the printed circuit board and partial distortion such as at screw tightening portions.
- Self heating (temperature increase) occurs when the power is turned ON, so the tolerance should be sufficient for the set thermal design.
- Carefully lay out the coil for the circuit board design of the non-magnetic shield type.
  A malfunction may occur due to magnetic interference.
- Use a wrist band to discharge static electricity in your body through the grounding wire.
- Do not expose the products to magnets or magnetic fields.
- Do not use for a purpose outside of the contents regulated in the delivery specifications.
- The products listed on this catalog are intended for use in general electronic equipment (AV equipment, telecommunications equipment, home appliances, amusement equipment, computer equipment, personal equipment, office equipment, measurement equipment, industrial robots) under a normal operation and use condition.
  The products are not designed or warranted to meet the requirements of the applications listed below, whose performance and/or quality require a more stringent level of safety or reliability, or whose failure, malfunction or trouble could cause serious damage to society, person or property.
  If you intend to use the products in the applications listed below or if you have special requirements exceeding the range or conditions set forth in the each catalog, please contact us.

(1) Aerospace/aviation equipment
(2) Transportation equipment (electric trains, ships, etc.)
(3) Medical equipment
(4) Power-generation control equipment
(5) Atomic energy-related equipment
(6) Seabed equipment
(7) Transportation control equipment
(8) Public information-processing equipment
(9) Military equipment
(10) Electric heating apparatus, burning equipment
(11) Disaster prevention/crime prevention equipment
(12) Safety equipment
(13) Other applications that are not considered general-purpose applications

When designing your equipment even for general-purpose applications, you are kindly requested to take into consideration securing protection circuit/device or providing backup circuits in your equipment.